

Project name : “Individual_Assignment_5”

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Course: HDS 5310

Project Purpose: To use linear regression to examine how opioid prescription rates and metro status predict access to syringe services programs (SSPs), measured by the cube root of distance. The assignment includes model building, assumption testing, and comparison using a Partial F-test.

Date last edited : 04-15-2025

```
#Load the libraries
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr       1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lmtest)
```

```
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
```

```
##
##      as.Date, as.Date.numeric
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##      recode
##
## The following object is masked from 'package:purrr':
##
##      some
```

STEP:1 Import the cleaned data set

```
# Load the data
ssp.data <- read.csv("C:/Users/drhud/Downloads/Individual Assignment 5/Individual Assignment 5/dist_ssp_amfar_ch9.csv")
```

```
summary(ssp.data)
```

```
##      county      STATEABBREVIATION      dist_SSP      HIVprevalence
## Length:500      Length:500      Min.   : 0.00      Min.   : -1.00
## Class :character Class :character 1st Qu.: 35.12      1st Qu.: 52.98
## Mode  :character Mode  :character Median : 75.94      Median : 101.15
##                                     Mean  :107.74      Mean   : 165.75
##                                     3rd Qu.:163.83      3rd Qu.: 210.35
##                                     Max.   :510.00      Max.   :2150.70
## opioid_RxRate      pctunins      metro
## Min.   : 0.20      Min.   : 3.00      Length:500
## 1st Qu.: 45.12      1st Qu.: 8.60      Class :character
## Median : 62.40      Median :11.70      Mode  :character
## Mean   : 68.33      Mean   :12.18
## 3rd Qu.: 89.95      3rd Qu.:15.00
## Max.   :345.10      Max.   :35.90
```

STEP 2 Create linear model with cube root transformation

```
# Create cube root transformation of distance to SSP
ssp.data <- ssp.data %>%
  mutate(cube_root_dist = dist_SSP^(1/3))
```

```
# Create linear regression model
dist.by.rx <- lm(cube_root_dist ~ opioid_RxRate, data = ssp.data)

# View model summary
summary(dist.by.rx)
```

```
##
## Call:
## lm(formula = cube_root_dist ~ opioid_RxRate, data = ssp.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2975 -1.0775 -0.0865  1.1314  3.5903
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.459643   0.133150  33.493  <2e-16 ***
## opioid_RxRate -0.001792   0.001716  -1.044   0.297
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.411 on 498 degrees of freedom
## Multiple R-squared:  0.002185,    Adjusted R-squared:  0.0001817
## F-statistic: 1.091 on 1 and 498 DF,  p-value: 0.2968
```

Interpretation:

The linear regression model assessed the relationship between opioid prescription rate and the cube root of distance to the nearest syringe services program (SSP). The coefficient for opioid_RxRate was -0.00179 with a p-value of 0.297, indicating that the predictor is not statistically significant at the 0.05 level.

The Adjusted R-squared was 0.00018, meaning the model explains less than 0.1% of the variation in SSP distance. The overall F-test was also not significant ($F(1, 498) = 1.091$, $p = 0.297$), suggesting that the model does not fit the data well.

Overall, this simple model does not provide evidence that opioid prescription rates are meaningfully associated with access to SSPs.

STEP 3 Checking assumptions

(Step 1) Independent observations

The data were collected at the county level across the U.S.

While counties are distinct units, some neighboring counties may share similar demographic, geographic, or healthcare system characteristics.

This geographic clustering could reduce true independence between observations.

=> This assumption was not met.

(Step 2) Continuous outcome

The outcome variable, distance to the nearest syringe services program (dist_SSP), is measured in miles.

It can take on a wide range of positive continuous values. => This assumption is met.

(Step 3) Linearity assumption

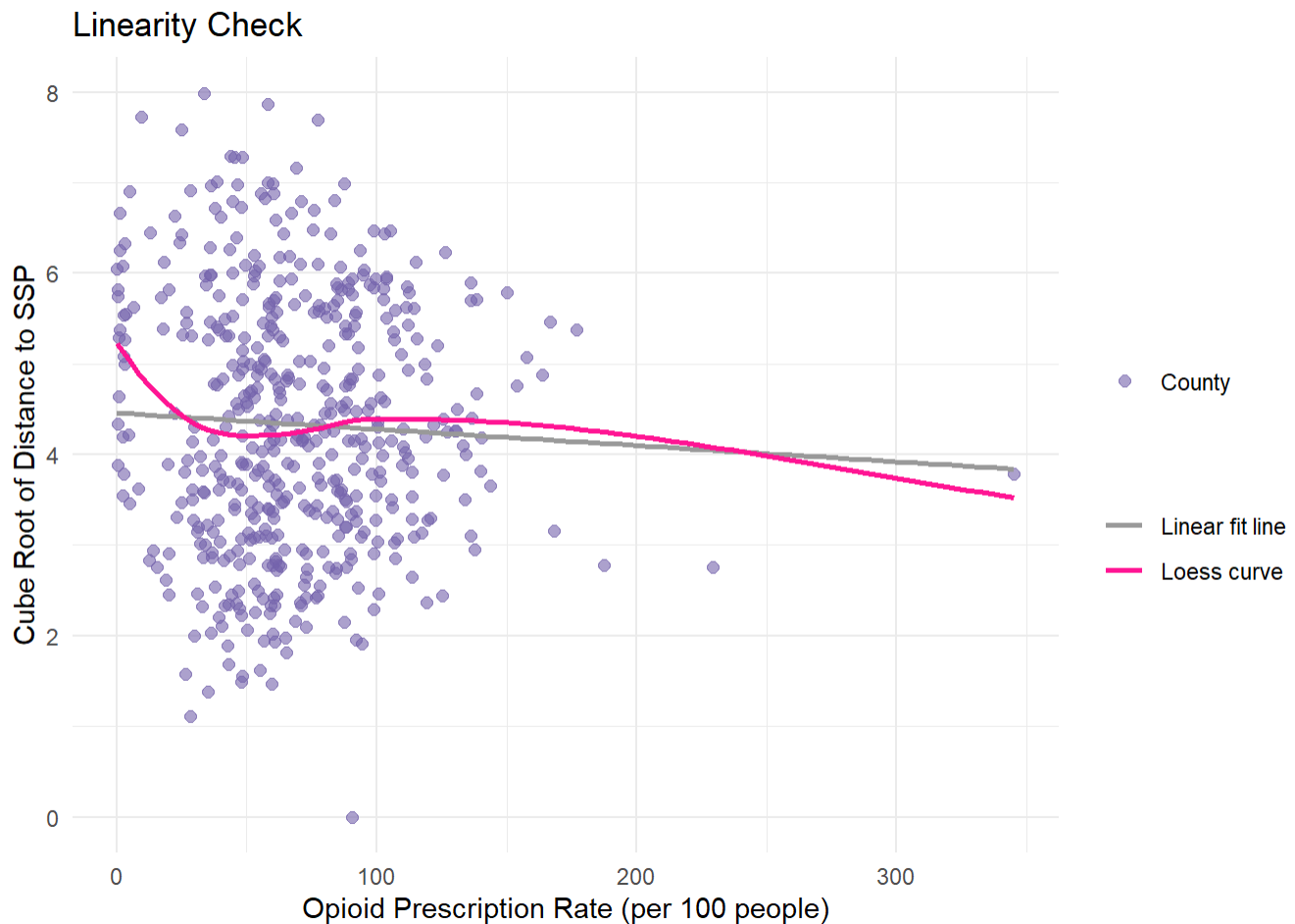
The linearity assumption is checked by creating a scatterplot of opioid_RxRate (predictor) and cube root of dist_SSP (outcome).

A linear regression line and a loess smoother help visualize the relationship.

If the loess curve follows the linear regression line closely, linearity is likely satisfied.

```
# Linearity check using ggplot with loess and linear regression line
ssp.data %>%
  ggplot(aes(x = opioid_RxRate, y = cube_root_dist)) +
  geom_point(aes(size = "County"), color = "#7463AC", alpha = 0.6) +
  geom_smooth(aes(color = "Linear fit line"), method = "lm", se = FALSE) +
  geom_smooth(aes(color = "Loess curve"), se = FALSE) +
  theme_minimal() +
  labs(
    title = "Linearity Check",
    x = "Opioid Prescription Rate (per 100 people)",
    y = "Cube Root of Distance to SSP"
  ) +
  scale_color_manual(values = c("gray60", "deeppink"), name = "") +
  scale_size_manual(values = 2, name = "")
```

```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Interpretation:

The Loess curve does not align closely with the linear regression line, especially at lower opioid prescription rates (under ~50 prescriptions per 100 people), where it shows a steep drop.

As the prescription rate increases, the Loess curve gradually flattens and runs mostly parallel to the linear fit, but some divergence remains across the predictor range.

This pattern suggests that the relationship between opioid prescription rate and the cube root of distance to SSP is not strictly linear, especially in the lower range. $\Rightarrow$ This assumption of linearity does not appear to be met.

(Step 4) Constant variance assumption

Breusch-Pagan test could be used to test the null hypothesis that the variance is constant.

Null hypothesis (H_0): The residuals have constant variance (homoscedasticity).

Alternative hypothesis (H_A): The residuals do not have constant variance .

```
# Testing for equal variance (homoscedasticity) using Breusch-Pagan test
const.var.test <- lmtest::bptest(formula = dist.by.rx)
const.var.test
```

```
##
## studentized Breusch-Pagan test
##
```

```
## data: dist.by.rx
## BP = 14.046, df = 1, p-value = 0.0001784
```

Interpretation:

The Breusch-Pagan test statistic has a tiny p-value (BP = 14.05; $p < .001$), which suggests that constant variance is not present. Because the p-value is less than 0.05, we reject the null hypothesis that the residuals have constant variance. This indicates that heteroscedasticity is present, and therefore, the assumption of homoscedasticity is not met in this model.

(Step 5) Independence of residuals

The Durbin-Watson test helps check this by measuring autocorrelation in the residuals.

The null hypothesis for the Durbin-Watson test is that the residuals are independent.

The alternative hypothesis is that the residuals are not independent.

```
# test independence of residuals
dw.test.rx <- lmtest::dwtest(formula = dist.by.rx)
dw.test.rx
```

```
##
## Durbin-Watson test
##
## data: dist.by.rx
## DW = 2.0081, p-value = 0.5356
## alternative hypothesis: true autocorrelation is greater than 0
```

Interpretation:

The D-W statistic is near 2 (DW = 2.0081) and the p-value was high ($p = 0.5356$), so we conclude that the null hypothesis is retained.

Since the null hypothesis was that the residuals were independent, we found that this assumption was met.

(Step 6) Normality of residuals

Normally distributed residuals indicate that the regression model explains the outcome variable consistently across its range.

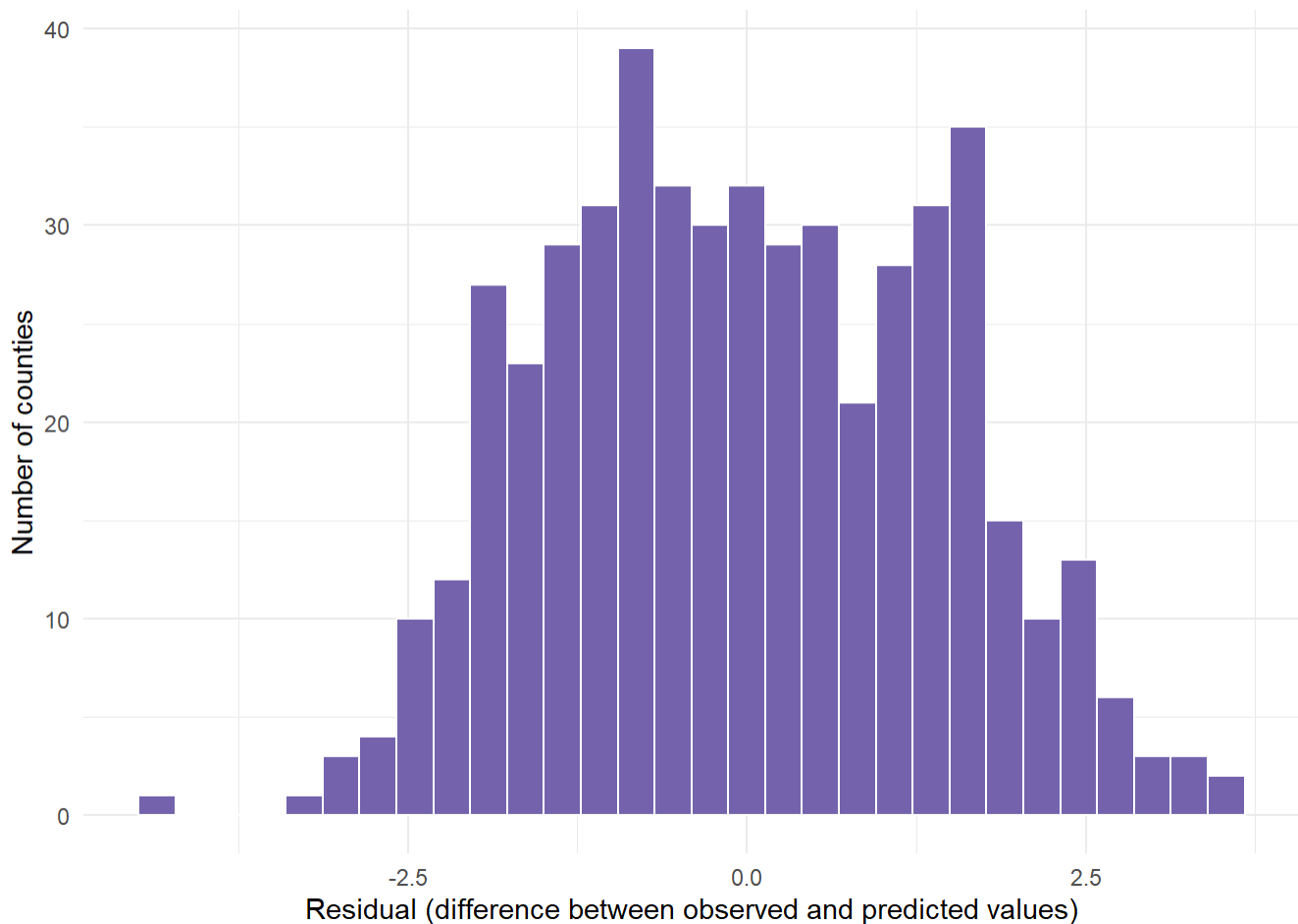
If the residuals are skewed, this suggests that the model performs better for some observations than others.

We check this assumption visually using a histogram of residuals.

```
# Step 6: Normality of residuals by plotting a histogram using ggplot()
data.frame(resid = dist.by.rx$residuals) %>%
  ggplot(aes(x = resid)) +
  geom_histogram(fill = "#7463AC", col = "white") +
  theme_minimal() +
  labs(
```

```
x = "Residual (difference between observed and predicted values)",
y = "Number of counties"
)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Interpretation:

The histogram of residuals appears approximately symmetric and bell-shaped, with no major skewness or outliers.

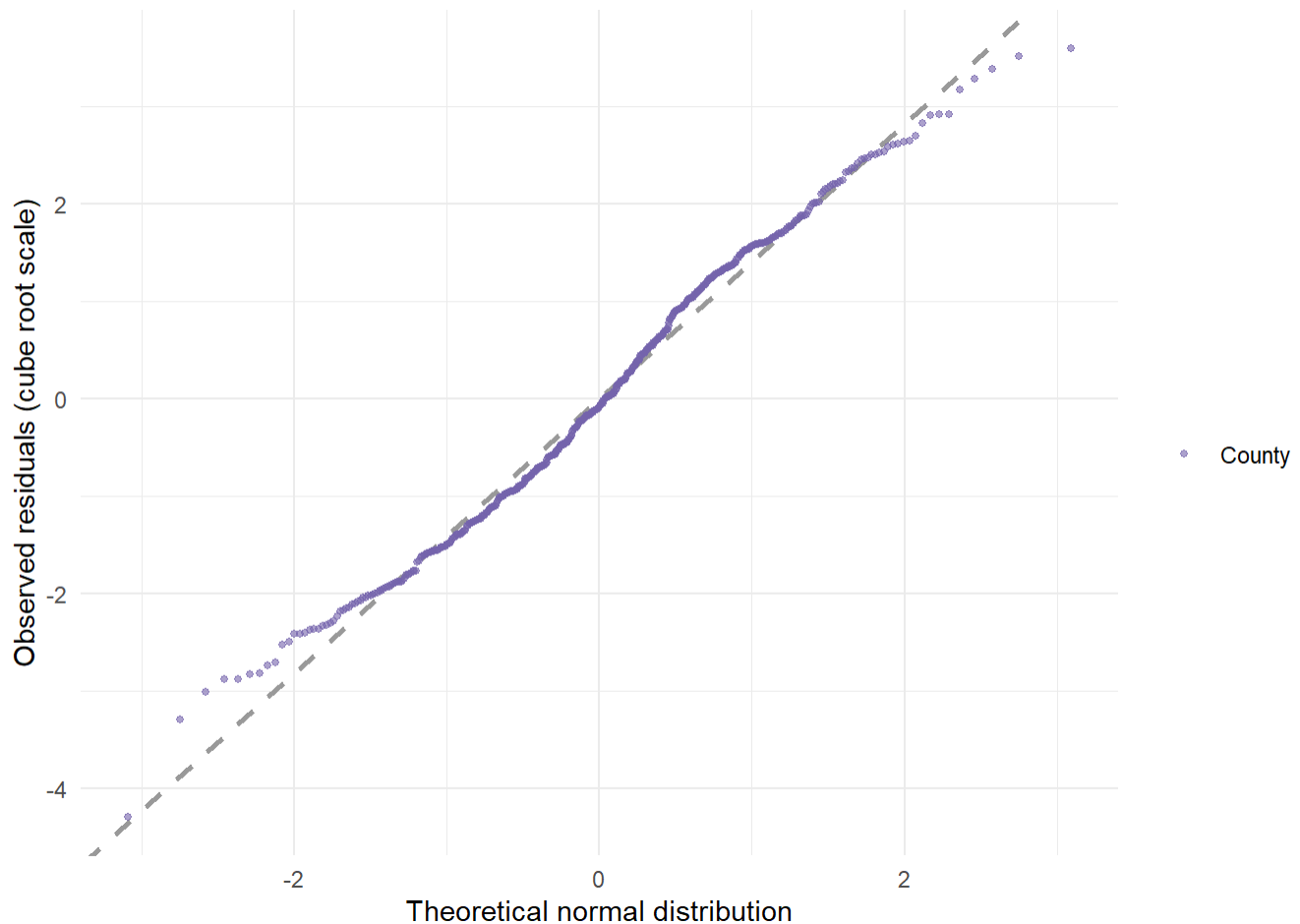
This suggests that the residuals are roughly normally distributed, and the assumption of normality is met.

Now Testing the normality of residuals assumption using QQ plot

```
#QQ plot
data.frame(resid = dist.by.rx$residuals) %>%
  ggplot(aes(sample = resid)) +
  geom_abline(
    aes(intercept = mean(resid), slope = sd(resid)),
    linetype = "dashed", color = "gray60", size = 1
  ) +
  stat_qq(aes(size = "County"), color = "#7463AC", alpha = 0.6) +
  theme_minimal() +
  labs(
    x = "Theoretical normal distribution",
```

```
y = "Observed residuals (cube root scale)"
) +
scale_size_manual(values = 1, name = "")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Interpretation

The Q-Q plot shows that the residuals closely follow the reference line, indicating that they are approximately normally distributed.

This suggests that the normality assumption is met.

STEP 4 Interpret the model results

```
summary(dist.by.rx)
```

```
##
## Call:
## lm(formula = cube_root_dist ~ opioid_RxRate, data = ssp.data)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2975 -1.0775 -0.0865  1.1314  3.5903
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.459643    0.133150  33.493  <2e-16 ***
## opioid_RxRate -0.001792    0.001716  -1.044    0.297
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.411 on 498 degrees of freedom
## Multiple R-squared:  0.002185,    Adjusted R-squared:  0.0001817
## F-statistic: 1.091 on 1 and 498 DF,  p-value: 0.2968
```

Interpreting the model results

The linear regression model was used to assess the relationship between opioid prescription rate and the cube root of distance to the nearest syringe services program (SSP).

The intercept was 4.46, indicating that when the opioid prescription rate is 0, the expected cube root of distance to an SSP is approximately 4.46 units.

The slope for opioid_RxRate was -0.00179, suggesting that for every one-unit increase in opioid prescriptions per 100 people, the cube root of distance to the nearest SSP decreases by 0.00179 units.

The p-value for the slope was 0.297, which is not statistically significant at the 0.05 level. This indicates that opioid prescription rate is not a significant predictor of the cube root of SSP distance in this model.

The R-squared value was 0.002, meaning that only 0.2% of the variance in the cube root of distance is explained by opioid prescription rate.

Overall, the model does not show a strong or statistically significant relationship and should be interpreted with caution, especially given the previously identified assumption violations.

Interpreting the results of the assumption checking

The linear regression analysis met some assumptions and did not meet others.

The assumptions it met were continuous outcome variable, independence of residuals, and approximate normality of residuals.

The assumptions it failed were independence of observations, linearity, and homoscedasticity.

Because the model does not meet all the assumptions, especially those involving structure and variance, it may be biased and should be interpreted cautiously.

STEP 5 Add the metro variable to the model

```
# Create a multiple linear regression model with opioid_RxRate and metro
dist.by.rx.met <- lm(cube_root_dist ~ opioid_RxRate + metro, data = ssp.data)

# View the updated model summary
summary(dist.by.rx.met)
```

```
##
## Call:
## lm(formula = cube_root_dist ~ opioid_RxRate + metro, data = ssp.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8170 -1.1021 -0.1212  1.0017  3.5949
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.052810    0.140077  28.933 < 2e-16 ***
## opioid_RxRate  -0.002606    0.001644  -1.585   0.114
## metronon-metro  0.843905    0.121476   6.947 1.17e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.348 on 497 degrees of freedom
## Multiple R-squared:  0.0905, Adjusted R-squared:  0.08684
## F-statistic: 24.73 on 2 and 497 DF, p-value: 5.781e-11
```

```
# View 95% confidence intervals
confint(dist.by.rx.met)
```

```
##              2.5 %      97.5 %
## (Intercept)  3.777593500 4.3280263991
## opioid_RxRate -0.005836199 0.0006240036
## metronon-metro 0.605235967 1.0825741881
```

STEP 6

Interpreting the model results

The multiple regression model included opioid prescription rate and metro status as predictors of the cube root of distance to the nearest syringe services program (SSP).

The p-value for opioid_RxRate was 0.114, which is not statistically significant, suggesting that opioid prescription rate does not significantly contribute to the model.

The p-value for metro was < .001, indicating that metro status is a statistically significant predictor of distance to an SSP.

The 95% confidence interval for the opioid_RxRate coefficient was [-0.0058, 0.00062], which includes 0 — confirming it is not statistically significant.

The 95% confidence interval for the metro coefficient was [0.6052, 1.0826], which does not include 0 — confirming its significance.

The model as a whole was statistically significant, with an F-statistic of 24.73 and a p-value of 5.78e-11.

The Adjusted R-squared was 0.08684, indicating that about 8.7% of the variation in the cube root of SSP distance is explained by this model.

This is a noticeable improvement from the simple model (Adjusted $R^2 = 0.00018$), which included only opioid prescription rate.

The coefficient for metro was 0.84, meaning that non-metro counties (reference = metro) are predicted to have a cube root SSP distance that is 0.84 units higher than metro counties, holding prescription rate constant.

Overall, adding metro status improved the model's explanatory power and revealed that urban-rural location plays a meaningful role in explaining access to SSPs.

STEP 7 Compare the larger and smaller models using the Partial-F test

```
# Compare the smaller and larger models using a Partial F-test
anova(dist.by.rx, dist.by.rx.metro)
```

```
## Analysis of Variance Table
##
## Model 1: cube_root_dist ~ opioid_RxRate
## Model 2: cube_root_dist ~ opioid_RxRate + metro
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      498 991.43
## 2      497 903.67   1    87.753 48.262 1.174e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

step 8 Interpret and report the results of the Partial-F test

Interpretation:

The Partial F-test produced an F-value of 48.26 with 1 and 497 degrees of freedom, and a p-value of 1.174e-11.

NHST Step-by-Step

Step 1: State the hypotheses

H_0 (null): The larger model (with metro) is no better than the smaller model at explaining the outcome.

H_1 (alt): The larger model is better than the smaller model at explaining the outcome.

Step 2: Compute the test statistic

The Partial F-value is 48.26 with 1 and 497 degrees of freedom.

Step 3: Assess the probability

The p-value is very small ($p = 1.174e-11$), which means the probability of observing this large an F-value by chance (if the null were true) is extremely low.

Step 4 & 5: Interpret the probability and write a conclusion

Since the p-value is much less than 0.05, we reject the null hypothesis.

The Partial F-test provides strong evidence that the larger model, which includes metro status, significantly improves the fit compared to the model with only opioid prescription rate.

Final Interpretation Paragraph

A linear regression model including opioid prescription rate and metro status of a county to explain the cube root of distance to the nearest syringe services program (SSP) was statistically significantly better than a baseline model including only opioid prescription rate [$F(1, 497) = 48.262$, $p = 1.174e-11$]. The model explained 8.684% of the variation in SSP distance (Adjusted $R^2 = 0.08684$).

While opioid prescription rate was not a statistically significant predictor of SSP distance ($b = -0.0020606$; $p = 0.114$), metro status was a statistically significant predictor ($b = 0.843905$; $p = 1.17e-11$). The 95% confidence interval for the metro coefficient ranged from 0.6052 to 1.0827, confirming statistical significance.

In this sample, non-metro counties were predicted to be 0.8439 units farther (in cube root distance) from the nearest syringe services program compared to metro counties, holding opioid prescription rate constant.

Overall, this suggests that rural status is a meaningful factor in access to syringe services programs. While the model explains only a modest portion of the variance, adding metro significantly improved model fit, and geographic factors like metro status should be considered in understanding access disparities.